

Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

- 1 Observer design on $SO(3)$
- 2 Gradient Observer design
- 3 Lyapunov analysis

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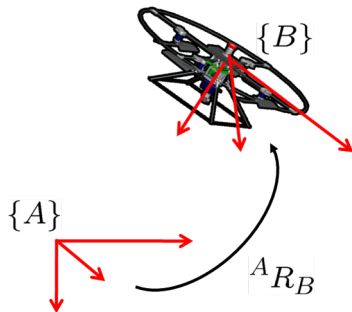
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Orientation between two frames

Attitude can be represented as rotation matrices in the special orthogonal group

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det(R) = 1\}$$

Alternative representations are $q \in \mathbb{R}^4$ unit quaternions, or as roll-pitch-yaw Euler angles (ψ, θ, ϕ) .



Attitude kinematics

Left invariant kinematics

$$\dot{X} = XU \quad (X = R \Rightarrow U = \Omega_{\times}), \quad \dot{R} = R\Omega_{\times}$$

$\Omega \in \{B\}$ is the rigid body angular velocity measured in the body-fixed-frame.

$$\Omega_{\times} = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix}$$

The angular velocity map $\times$ is the matrix representation of the vector product. That is $\Omega_{\times} v = \Omega \times v$ for any vector $v \in \mathbb{R}^3$.

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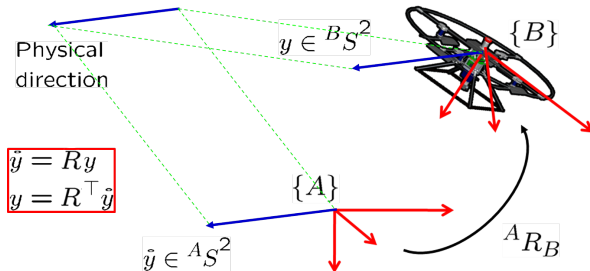
Motivation: Attitude estimation for UAVs

A sensor that measures the orientation in 3D space does not exist!

- A good quality, high bandwidth estimate of attitude is required for control of aerial robotic vehicles.
- Estimation algorithm must be able to run effectively on embedded hardware.
- It must deal robustly with poor quality noisy measurements obtained from low-cost light-weight inertial measurement units.

Observer based on vector direction measurements

The measurement of the orientation matrix is obtained by means of the measurement in a moving frame of known directions in the inertial frame.



Both y and $\hat{y}$ must be measured or known to obtain information on R .

Observer based on vector direction measurements

For example, magnetic field, gravitational field, far field target bearings on the sun, horizon, etc.

Known vector direction $\mathring{y} \in S^2$, expressed in $\{A\}$.

The measurement is made in the body-fixed frame $\{B\}$

$$y_i = R^\top \mathring{y}_i, \quad i = \{1 \dots n\}$$

Observer based on vector direction measurements

System

$$\dot{R} = R\Omega_{\times}, \quad R \in SO(3), \quad \Omega_{\times} \in \mathfrak{so}(3)$$

Output for known 'references' $\dot{y}_i \in S^2$, $i = \{1 \dots n\}$

$$y_i = h(R, \dot{y}_i) = R^{\top} \dot{y}_i$$

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Output system dynamics

$$\dot{x}_i = -\Omega \times x_i, \quad \text{the measurment } y_i = x_i$$

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Two options to design observers

- ① Design an observer of each Output system, and then solve an algebraic problem in order to build $\hat{R}$,
- ② Design the observer of the Group system.

Observer of each output system

System kinematics

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Observer on the output system

$$\dot{\hat{x}}_i = -\Omega_{\times} \hat{x} + \beta(y_i, \hat{y}_i), \quad \hat{y}_i = \hat{x}_i$$

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In order to preserve the norm of $\hat{x}$ ($\hat{x} \in S^2$), the innovation $\beta = k_P \alpha(y_i, \hat{y}_i)_{\times} \hat{x}_i$,

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building an estimate on SO(3)

From the estimates $\hat{x}_i$, $i = \{1 \dots N\}$, build the estimate on $SO(3)$.

Building an estimate on $SO(3)$

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Suppose the measurement of two noncollinear directions $\hat{y}_1$ and $\hat{y}_2$ are available. In that case, one verifies that y_1 and y_2 are noncollinear, but nothing can prevent that $\hat{y}_1$ becomes collinear to $\hat{y}_2$ and hence it is not possible to ensure that one can build an $\hat{R}$ during the transient.

Observer of the group system directly

Consider a system and an observer on $SO(3)$

$$\dot{R} = R\Omega_{\times}, y_i = R^T \dot{y}_i$$

Build an observer on $SO(3)$,

$$\dot{\hat{R}} = \hat{R} (\Omega + \alpha(y_i, \hat{y}_i))_{\times}$$

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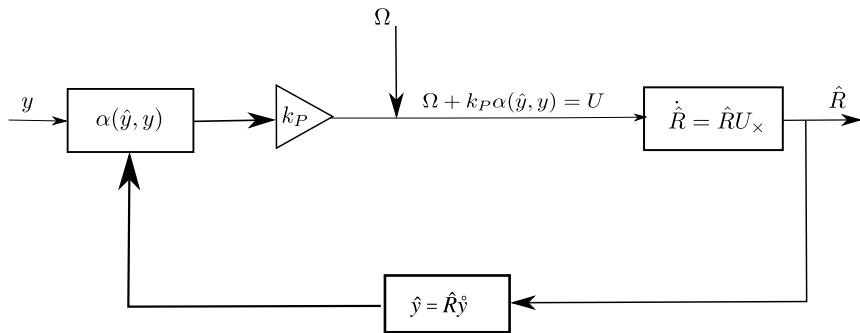
Build an observer on $SO(3)$,

$$\dot{\hat{R}} = \hat{R}(\Omega + \alpha(y_i, \hat{y}_i))_{\times}$$

The observer has an **internal model** of the system (identical initial conditions lead to identical trajectories), more importantly the observer is synchronous to the original system with respect to the error $E_1 = \hat{R}R^T$.

The observer error dynamics are driven purely by the innovation term α .

Block Diagram of Observer



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Gradient Observer design for $SO(3)$

Group system is given by

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Structure of the observer

$$\dot{\hat{R}} = \hat{R}(\Omega + k_P \alpha(y_i, \hat{y}_i))_{\times}$$

$$\hat{y}_i = \hat{R}^{\top} \dot{y}_i$$

Gradient Observer design for $SO(3)$

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Structure of the observer

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The output cost is based on the chordal distance on the sphere $S^2 \in \mathbb{R}^3$,

$$\ell_{\dot{y}_i}(\hat{R}, y_i) = \frac{1}{2} \|\hat{R}y_i - \dot{y}_i\|^2.$$

Vectorial directions cost function

Gradient observer design:

$$\dot{\hat{R}} = \hat{R}\Omega_{\times} - k_P \text{grad}_{\hat{R}} \sum_{i=1}^N k_i \ell_{\dot{y}}(\hat{R}, y_i)$$

$$\ell_{\dot{y}}(\hat{R}, y_i) = \frac{1}{2} \|\hat{R}y_i - \dot{y}_i\|^2.$$

The invariance of the cost is a consequence of the invariance of the Euclidean norm operator with respect to rotation:

$$\frac{1}{2} \|Q\hat{R}y_i - Q\dot{y}_i\|^2 = \frac{1}{2} \|Q(\hat{R}y_i - \dot{y}_i)\|^2 = \frac{1}{2} \|\hat{R}y_i - \dot{y}_i\|^2, \quad Q \in SO(3)$$

Vectorial directions cost function

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Known that $y_i = R^{\top} \dot{y}_i$,

$$\ell_{\dot{y}}(\hat{R}, y_i) = \frac{1}{2} \|\hat{R}R^{\top} \dot{y}_i - \dot{y}_i\|^2 = \frac{1}{2} \|E_1 \dot{y}_i - \dot{y}_i\|^2.$$

Gradient of $\ell_{\hat{y}_i}(\hat{R}, y_i)$

Recall that:

$$\ell_{\hat{y}_i}(\hat{R}, y_i) = \frac{1}{2} \|\hat{R}y_i - \hat{y}_i\|^2 = \frac{1}{2} \text{tr} \left((\hat{R}y_i - \hat{y}_i)(\hat{R}y_i - \hat{y}_i)^\top \right)$$

one has:

$$\begin{aligned} D_{\hat{R}} \ell_{\hat{y}_i}(\hat{R}, y_i) [\Delta_\times \hat{R}] &= \text{tr} \left((\hat{R}y_i - \hat{y}_i)(\Delta_\times \hat{R}y_i)^\top \right) \\ &= -\langle \mathbb{P}_{\mathfrak{so}(3)} \left((\hat{R}y_i - \hat{y}_i)y_i^\top \hat{R}^\top \right) \hat{R}, \Delta_\times \hat{R} \rangle \end{aligned}$$

and hence:

$$\text{grad}_{\hat{R}} \ell_{\hat{y}_i}(\hat{R}, y_i) = -\mathbb{P}_{\mathfrak{so}(3)} \left((\hat{R}y_i - \hat{y}_i)y_i^\top \hat{R}^\top \right) \hat{R} = -\frac{1}{2} \hat{R}(y_i \times \hat{y}_i)_\times$$

From there: $\alpha(y, \hat{y}) = \frac{1}{2} \sum_{i=1}^N k_i (y_i \times \hat{y}_i)$

Vectorial directions cost function

The minimum of the cost function is obtained when $e_i^l \equiv \dot{y}_i$
 ($e_i^l = E_l \dot{y}_i$),

If the cost function is a collection of cost functions associated to at least two linearly independent inertial directions:

$$\mathcal{L} = \sum_{i=1}^N k_i \ell_{\dot{y}_i}(\hat{R}, y_i) = \sum_{i=1}^N \frac{k_i}{2} \|E_l \dot{y}_i - \dot{y}_i\|^2, \quad k_i > 0$$

it is straightforward to verify that the minimum $\mathcal{L} = 0$ is obtained when:

$$y_i \equiv \dot{y}_i, \quad \forall i = \{1 \dots N\} \Rightarrow E_l \equiv I \Rightarrow \hat{R} \equiv R$$

If $\Omega_y = \Omega$ (no bias), the cost function $\mathcal{L}$ can be used as a Lyapunov function candidate.

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Lyapunov analysis

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$$\dot{\mathcal{L}} = \sum_{i=1}^N k_i (E_1 \dot{y}_i - \dot{y}_i)^\top \dot{E}_1 \dot{y}_i$$

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Recall that $E_i = \hat{R} R^\top$,

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$$\begin{aligned} \dot{E}_i &= \dot{\hat{R}} R^\top + \hat{R} \dot{R}^\top \\ &= \hat{R}(\Omega + k_P \alpha(y_i, \hat{y}_i))_\times R^\top - \hat{R} \Omega_\times R^\top \\ &= k_P \hat{R} \alpha(y_i, \hat{y}_i)_\times R^\top = k_P (\hat{R} \alpha(y_i, \hat{y}_i)_\times \hat{R}^\top) \hat{R} R^\top \\ &= k_P \text{Ad}_{\hat{R}} \alpha(y_i, \hat{y}_i)_\times E_i \end{aligned}$$

Lyapunov analysis

$$\dot{\mathcal{L}} = \sum_{i=1}^N k_i (E_l \dot{y}_i - \dot{\hat{y}}_i)^\top \dot{E}_l \dot{y}_i, \quad \dot{E}_l = k_P \text{Ad}_{\hat{R}} \alpha(y_i, \hat{y}_i)_{\times} E_l$$

This implies

$$\begin{aligned} \dot{\mathcal{L}} &= k_P \sum_{i=1}^N k_i (E_l \dot{y}_i - \dot{\hat{y}}_i)^\top \text{Ad}_{\hat{R}} \alpha(y_i, \hat{y}_i)_{\times} E_l \dot{y}_i, \\ &= k_P \sum_{i=1}^N k_i \text{tr} \left(E_l \dot{y}_i (E_l \dot{y}_i - \dot{\hat{y}}_i)^\top \text{Ad}_{\hat{R}} \alpha(y_i, \hat{y}_i)_{\times} \right) \end{aligned}$$

Define the orthogonal projection onto the skew-symmetric matrices $\mathbb{P}_{\mathfrak{so}(3)}: \mathbb{R}^{3 \times 3} \rightarrow \mathfrak{so}(3)$, by

$$\mathbb{P}_{\mathfrak{so}(3)}(Z) := \frac{1}{2}(Z - Z^\top).$$

Lyapunov analysis

Since $\alpha_{\times} \in \mathfrak{so}(3)$ (and therefore $\text{Ad}_{\hat{R}} \alpha_{\times} \in \mathfrak{so}(3)$) then

$$\begin{aligned}
 \dot{\mathcal{L}} &= k_P \sum_{i=1}^N k_i \text{tr} \left((E_l \dot{y}_i \dot{y}_i^{\top} - \dot{y}_i \dot{y}_i^{\top} E_l^{\top}) \text{Ad}_{\hat{R}} \alpha(y_i, \hat{y}_i)_{\times} \right) \\
 &= k_P \sum_{i=1}^N k_i \text{tr} \left((E_l \dot{y}_i \dot{y}_i^{\top} - \dot{y}_i \dot{y}_i^{\top} E_l^{\top}) \hat{R} \alpha(y_i, \hat{y}_i)_{\times} \hat{R}^{\top} \right) \\
 &= k_P \sum_{i=1}^N k_i \text{tr} \left((\hat{R} R^{\top} \dot{y}_i \dot{y}_i^{\top} - \dot{y}_i \dot{y}_i^{\top} R \hat{R}^{\top}) \hat{R} \alpha(y_i, \hat{y}_i)_{\times} \hat{R}^{\top} \right) \\
 &= k_P \sum_{i=1}^N k_i \text{tr} \left(\hat{R}^{\top} (\hat{R} R^{\top} \dot{y}_i \dot{y}_i^{\top} - \dot{y}_i \dot{y}_i^{\top} R \hat{R}^{\top}) \hat{R} \alpha(y_i, \hat{y}_i)_{\times} \right) \\
 &= k_P \text{tr} \left(\sum_{i=1}^N k_i (y_i \hat{y}_i^{\top} - \hat{y}_i y_i^{\top}) \alpha(y_i, \hat{y}_i)_{\times} \right)
 \end{aligned}$$

Lyapunov analysis

Using

$$\alpha(y_i, \hat{y}_i)_{\times} = \frac{1}{2} \sum_{i=1}^N k_i (y_i \times \hat{y}_i)_{\times} = \frac{1}{2} \sum_{i=1}^N k_i (y_i \hat{y}_i^{\top} - \hat{y}_i y_i^{\top})^{\top},$$

One has

$$\dot{\mathcal{L}} = -\frac{k_P}{2} \left\| \sum_{i=1}^N k_i (y_i \times \hat{y}_i) \right\|^2$$

This function is negative as long as $E_1 \dot{M} = \dot{M} E_1^{\top}$
 ($\dot{M} = \sum_{i=1}^N k_i \dot{y}_i \dot{y}_i^{\top}$) or equivalently $E_1 \neq E_1^{\top}$

Lyapunov analysis for a single output system

Structure of the observer

$$\dot{y}_i = -\Omega_{\times} y_i$$

$$\dot{\hat{y}}_i = -(\Omega - k_P \alpha(y_i, \hat{y}_i))_{\times} \hat{y}_i$$

The output cost is based on the chordal distance on the sphere
 $S^2 \in \mathbb{R}^3$,

$$\begin{aligned} \ell_{\hat{y}_i}(\hat{R}, y_i) &= \frac{1}{2} \|y_i - \hat{y}_i\|^2, \\ &= 1 - y_i^{\top} \hat{y}_i \end{aligned}$$

Lyapunov analysis for a single output system

The cost function can be used as a Lyapunov function candidate for a single output system ($\mathcal{L} \equiv \ell_{\hat{y}_i}(\hat{R}, y_i)$)

$$\begin{aligned}\dot{\mathcal{L}} &= -\dot{y}_i^\top \hat{y}_i - y_i^\top \dot{\hat{y}}_i \\ &= y_i^\top \Omega_\times \hat{y}_i - y_i^\top \Omega_\times \hat{y}_i - k_P y_i^\top \alpha(y_i, \hat{y}_i)_\times \hat{y}_i \\ &= +k_P y_i^\top (\hat{y}_i)_\times \alpha(y_i, \hat{y}_i) = +k_P (y_i \times \hat{y}_i)^\top \alpha(y_i, \hat{y}_i)\end{aligned}$$

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Recalling that $\alpha = -\frac{1}{2}(y_i \times \hat{y}_i)$, $\Rightarrow \dot{\mathcal{L}} = -\frac{k_P}{2} \|y_i \times \hat{y}_i\|^2$.

This function is negative as long as y_i is not parallel to $\hat{y}_i$

